

TAHMEED UR RAWFUN

Civil and Structural Engineering Student | Computational Design & Data Analysis

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PROFESSIONAL SUMMARY

Civil & Environmental Engineering graduate from the Islamic University of Technology (IUT), majoring in Structural Engineering. Skilled in structural analysis, finite element methods, reinforced concrete and steel design, and structural modelling using ETABS, SAP2000, SAFE, Revit, and AutoCAD. Experienced in developing Python-based engineering tools for structural design, wind-load analysis, finite element analysis, and structural optimization, with a strong interest in computational structural engineering.

TECHNICAL SKILLS

- **STRUCTURAL ENGINEERING** : Finite Element Method (FEM), Matrix Structural Analysis, Reinforced Concrete Design, Steel Design, Structural Dynamics, Seismic Analysis, Wind Load Analysis, Load Combinations
- **DESIGN CODES** : BNBC 2020, ACI 318-19, ASCE 7-16/22 , AISC Specifications
- **STRUCTURAL SOFTWARE & BIM** : ETABS, SAP2000, SAFE, Autodesk Revit, STAAD.Pro, AutoCAD
- **PROGRAMMING & COMPUTATIONAL** : Python (NumPy, Pandas, Matplotlib), Streamlit, MATLAB, C/C++, Git, HTML/CSS, JavaScript

RELEVANT PROJECTS & ENGINEERING EXPERIENCE

- **AutoRC-Gen (Automated Reinforced Concrete Design Tool)** : *Python, Streamlit, ACI 318*
 - ✓ Architected an interactive web application automating reinforced concrete member sizing and reinforcement detailing, reducing calculation time and eliminating manual design errors
 - ✓ Executed automated code-compliant structural checks for flexure, shear, and serviceability limits in beams and columns
- **Automated Wind Load & Pressure Calculator** : *Python, Streamlit, BNBC 2020*
 - ✓ Developed a web-based computational tool automating design wind pressure and lateral load distributions for multi-storey buildings in accordance with height profiles.
 - ✓ Automated velocity pressure coefficients, gust effect factor, and directional factor calculations across variable exposure categories and height profiles.
- **Multi-Storey Residential Building Design and Analysis** : *ETABS, AutoCAD*
 - ✓ Executed comprehensive 3D building structural modeling with customized storey heights, floor diaphragms, and soil classifications, ensuring accurate gravity and lateral load distribution
- **StructPredict (ML-Based Structural Engineering Tool)** : *Python, Scikit-Learn, Pandas*
 - ✓ Developed a python application using a Random Forest surrogate model to predict reinforced-concrete column axial and flexural capacities.
 - ✓ Integrated geometric, material, and reinforcement parameters into an interactive prediction interface for rapid structural capacity estimation.

EDUCATION

- BSc in Civil and Environmental Engineering (Expected 2026)
 - Islamic University of Technology (IUT) – Cumulative GPA : 3.11 / 4.00
- Higher Secondary Certificate (HSC), GPA : 5.00/5.00 – Dhaka College, Dhaka
- Secondary School Certificate (SSC), GPA : 5.00/5.00 – Birshrestha Noor Mohammad Public College, Dhaka

LANGUAGES

- **Bangla** : Native
- **English** : Professional Working
- **Hindi** : Professional Working